BacSense

Version 2.0

Technical Architecture & Model Documentation

Cascaded Hybrid Classifier for Waterborne Bacterial Identification

VGG16 Transfer Learning + Hand-Crafted Feature Engineering + RBF-SVM

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1. Executive Summary

BacSense v2 is a two-stage cascaded hybrid classifier for the identification of five waterborne bacterial species from Gram-stained microscopy images. The system combines deep transfer learning (VGG16) with rich hand-crafted feature engineering and a binary specialist Support Vector Machine to resolve a critical confusion pair that the original single-stage model completely failed on.

Problem: The original BacSense v1 achieved 95.83% overall accuracy but had near-zero F1 scores for Escherichia coli and Pseudomonas aeruginosa — two morphologically similar Gram-negative rods that VGG16 FC features alone could not separate.

Solution: A cascaded architecture that routes ambiguous predictions to a specialist binary SVM trained on a 683-dimensional feature vector combining VGG16 deep features with seven hand-crafted descriptors targeting the discriminative signals invisible to the main model.

Metric	v1 (Main SVM only)	v2 (Cascaded)
Overall Accuracy	95.83%	95.65%
E. coli F1	~0.00 (failing)	0.9568
P. aeruginosa F1	~0.00 (failing)	0.9563
P. aeruginosa Recall	~0.00	0.9480
Specialist ROC-AUC	—	0.9863
CV F1 (Specialist)	—	0.9498

2. Dataset

2.1 Source: DIBaS (Digital Image of Bacteria Species)

The DIBaS dataset, introduced by Zielinski et al. (2017) in PLOS ONE, is the primary source of microscopy images. It contains 660 images across 33 bacterial species captured at 100x magnification with oil immersion. BacSense uses a 5-class subset of 307 original images corresponding to clinically relevant waterborne pathogens.

Species	Gram	Shape	Original Images	Risk Level
Escherichia coli	Negative	Rod	59	High
Pseudomonas aeruginosa	Negative	Rod	63	High
Enterococcus faecalis	Positive	Coccus	62	Medium
Clostridium perfringens	Positive	Rod	61	High
Listeria monocytogenes	Positive	Rod	62	High

2.2 Data Augmentation

The original 307 images are insufficient for robust deep feature training. An aggressive augmentation pipeline expands the dataset 18.7x to approximately 5,000 images for main model training. For specialist training, each of the two confusion classes is independently augmented to 800 images.

Parameter	Value
Rotation range	360 degrees (full circular)
Width / height shift	10%
Horizontal / vertical flip	Enabled
Zoom range	15%
Brightness range	0.85 – 1.15
Fill mode	Reflect
Target per class (specialist)	800 augmented + original PNG

For the specialist, original PNG images (59 E. coli + 63 P. aeruginosa) are combined with augmented JPG images during training to prevent domain shift artifacts between file formats. Total specialist training data: 1,722 images.

3. System Architecture

3.1 Overview: Two-Stage Cascaded Design

BacSense v2 uses a cascaded architecture where a fast 5-class main SVM provides an initial prediction. If that prediction falls within the known confusion pair (E. coli or P. aeruginosa), the image is routed to a dedicated specialist binary SVM that uses a much richer feature representation to make the final call.

Architecture Flow: Input Image → VGG16 Feature Extractor → PCA(94) → Main SVM → [if ambiguous] → 683-dim Feature Extraction → PCA(243) → Specialist SVM → Final Prediction

	Stage 1: Main SVM	If Ambiguous	Stage 2: Specialist	Final Output
Input Raw image	**Features** VGG16 (512-dim) → PCA (94-dim)	**Trigger** E. coli or P. aeruginosa predicted	**Features** 683-dim rich vector → PCA (243-dim)	**Output** 5-class or binary species label

3.2 Stage 1: Main Classifier

3.2.1 Feature Extractor — VGG16

VGG16 is used as a frozen feature extractor with ImageNet pre-trained weights. The top classification layer is removed, and the output of the last global average pooling layer provides a 512-dimensional feature vector for each input image. Images are resized to 128 x 128 pixels and normalized to [0, 1] before passing through the network.

Component	Detail
Base architecture	VGG16 (Simonyan & Zisserman, 2014)
Pre-training	ImageNet (1.2M images, 1000 classes)
Fine-tuning	None — weights fully frozen
Input size	128 x 128 x 3 (RGB)
Output	512-dimensional feature vector

3.2.2 Dimensionality Reduction — PCA

Principal Component Analysis reduces the 512-dimensional VGG16 output to 94 components while retaining 95% of explained variance. This reduces overfitting risk, lowers SVM training time, and removes correlated feature dimensions that introduce noise into the decision boundary.

3.2.3 Main Classifier — RBF-SVM

A Radial Basis Function Support Vector Machine is trained on the PCA-reduced features for 5-class classification. The SVM uses a one-vs-one decision strategy with class-balanced weighting to handle slight class imbalance in the original dataset.

Hyperparameter	Value	Selection Method
Kernel	RBF	Fixed — standard for high-dim features
C (regularization)	10	GridSearchCV, 3-fold CV
Gamma	0.01	GridSearchCV, 3-fold CV
Class weight	Balanced	Fixed — handles class imbalance
Decision function	OVO (one-vs-one)	Default for multi-class SVC

4. Stage 2: Specialist Classifier

4.1 Motivation — Why a Specialist?

E. coli and P. aeruginosa are both Gram-negative, rod-shaped bacteria with similar cell dimensions and staining characteristics. The standard VGG16 FC feature vector collapses their representations into overlapping regions in feature space, producing near-zero F1 scores for both species in the main model. Three peer-reviewed papers converge on the same solution: richer feature concatenation targeting color distribution, texture micropatterns, and spatial arrangement.

Paper	Key Finding	Feature Applied in BacSense
Zielinski et al., 2017 (PLOS ONE)	Color distribution features explicitly recommended for morphologically similar species	HSV histogram (48-dim)
Wahid et al. (Inception-v3 + SVM)	Feature concatenation is the principled fix for intra-species confusion	Full 683-dim concatenated vector
Rachmad et al., 2020	Binary CNN+SVM specialist viable for rod-shaped bacteria	Cascaded binary specialist SVM

4.2 Feature Vector — 683 Dimensions

The specialist uses a 683-dimensional feature vector formed by concatenating VGG16 deep features with seven hand-crafted descriptors. Each descriptor targets a specific visual property that differentiates E. coli from P. aeruginosa in Gram-stained images.

Feature Group	Dims	What It Captures	Target Signal
VGG16 FC Features	512	Deep semantic representation from ImageNet	General shape, high-level patterns
HSV Histogram	48	Color distribution across Hue/Sat/Value bins (4x4x3)	Stain intensity differences
Morphological	5	Area, perimeter, circularity, aspect ratio, solidity	Cell shape and size
LBP Histogram	59	Local Binary Patterns, P=8, R=1, uniform	Texture micropatterns, biofilm
GLCM Descriptors	24	Contrast, dissimilarity, homogeneity, energy, correlation, ASM (4 angles)	Spatial gray-level co-occurrence
Channel Stats	9	Mean, std, skewness per H/S/V channel	Fine-grained color statistics
Density Grid	16	4x4 spatial grid of foreground pixel density	Clustering vs dispersal pattern

Hu Moments	7	7 rotation-invariant moment descriptors (log-transformed)	Global shape invariants
Curvature	3	Mean, std, max curvature across all contour points	Rod bending — key P. aeruginosa signal

4.3 Why Each Feature Matters for This Pair

LBP (Local Binary Patterns)

P. aeruginosa is known to form loose biofilm-like clusters under microscopy. LBP captures the local texture microstructure created by these aggregations — a signal completely absent from VGG16 FC features which operate at a semantic rather than textural level.

GLCM (Gray-Level Co-occurrence Matrix)

GLCM descriptors computed at four angles (0, 45, 90, 135 degrees) capture the spatial relationship between intensity values across the image. P. aeruginosa typically shows higher local contrast and dissimilarity values due to its clustering behavior and slightly different staining depth.

Curvature Statistics

P. aeruginosa rods tend to exhibit more curvature and bending than E. coli rods, which are typically straighter. Curvature is computed as the cross product of consecutive edge vectors along detected contours, summarized as mean, standard deviation, and maximum curvature values across all contours in the image.

Spatial Density Grid

A 4x4 grid divides the image into 16 cells and computes foreground pixel density in each cell. This captures the spatial distribution and clustering pattern of bacteria across the slide — P. aeruginosa tends to form denser local aggregations while E. coli distributes more uniformly.

4.4 Specialist Pipeline

Step	Operation	Output Dims
1. VGG16 extraction	Forward pass through frozen VGG16	512
2. Hand-craft extraction	HSV + Morph + LBP + GLCM + ChannelStats + Density + Hu + Curvature	171
3. Concatenation	np.concatenate([vgg, rich])	683
4. Standardization	StandardScaler (zero mean, unit variance)	683
5. PCA	Retain 95% variance	243

6. SVM prediction	RBF-SVM, C=100, gamma=0.001, probability=True	Binary (0=E.coli, 1=P.aeru)
7. Confidence gate	Accept specialist if prob >= 0.90, else fallback to Stage 1	Final label

4.5 Confidence Threshold Gate

A confidence gate prevents the specialist from overriding the main SVM when it is uncertain. If the specialist's maximum class probability is below 0.90, the main SVM's prediction is used as the final answer. This is a practical engineering decision that prevents low-confidence specialist predictions from overriding a correct main SVM result, at the cost of occasionally retaining main SVM errors in the E. coli / P. aeruginosa pair.

Design Note: The 0.90 threshold was chosen based on observed confidence distributions. Future work should calibrate this threshold using a held-out validation set to maximize the F1 score of the gated system.

5. Training Details

5.1 Main Model Training

Component	Configuration
Training images	~5,000 (307 original × 18.7x augmentation)
PCA components	94 (95% variance retained from 512-dim input)
SVM kernel	RBF
Hyperparameter search	GridSearchCV, 3-fold cross-validation
Search space C	[0.1, 1, 10, 100]
Search space gamma	["scale", 0.1, 0.01, 0.001]
Best params	C=10, gamma=0.01
Scoring metric	F1 (macro)
Test accuracy	95.83%

5.2 Specialist Model Training

Component	Configuration
Training images	1,722 total (800 aug E.coli + 800 aug P.aeru + 122 original PNG)
Feature vector	683-dim (512 VGG16 + 171 hand-crafted)
PCA components	243 (95% variance retained from 683-dim input)
SVM kernel	RBF
Hyperparameter search	GridSearchCV, 3-fold cross-validation
Search space C	[1, 10, 100]
Search space gamma	["scale", 0.01, 0.001]
Best params	C=100, gamma=0.001
Class weight	Balanced
Scoring metric	F1 (binary)
CV F1	0.9498
Test accuracy	95.65%

5.3 Domain Shift Handling

The augmented training images were saved as JPEG files, while the original DIBaS images are PNG. JPEG compression introduces subtle color and texture artifacts that shift the feature

distribution, causing the specialist (trained on augmented JPEGs) to misclassify original PNG images at inference. This was resolved by mixing all original PNG images into the specialist training set, forcing the model to learn features robust to both formats.

6. Evaluation Results

6.1 Specialist Classifier Performance

Metric	E. coli	P. aeruginosa	Macro Avg
Precision	0.9486	0.9647	0.9566
Recall	0.9651	0.9480	0.9565
F1-Score	0.9568	0.9563	0.9565
Support	172	173	345

Overall test accuracy: 95.65% on 345 held-out images. ROC-AUC: 0.9863, indicating strong probability calibration across the decision boundary.

6.2 Improvement Over v1

Metric	v1 Main Model	v2 Specialist	Improvement
E. coli F1	~0.00	0.9568	+0.9568
P. aeruginosa F1	~0.00	0.9563	+0.9563
P. aeruginosa Recall	~0.00	0.9480	+0.9480
CV F1	N/A	0.9498	—
ROC-AUC	N/A	0.9863	—

6.3 Integration Test

End-to-end testing of the full cascaded pipeline on original held-out images (not seen during training) produced the following results:

Test	Correct / Total	Notes
E. coli (original PNG)	5 / 5	All routed to specialist, correctly classified
P. aeruginosa (original PNG)	5 / 5	All routed to specialist, correctly classified
Total	10 / 10	100% on held-out originals

7. Technical Stack

Component	Library / Version	Purpose
Deep feature extractor	TensorFlow / Keras + VGG16	ImageNet transfer learning
Dimensionality reduction	scikit-learn PCA	Variance-preserving compression
Main classifier	scikit-learn SVC (RBF)	5-class prediction
Specialist classifier	scikit-learn SVC (RBF, probability=True)	Binary E.coli / P.aeru
LBP features	scikit-image local_binary_pattern	Texture micropatterns
GLCM features	scikit-image graycomatrix / graycoprops	Spatial co-occurrence
Color / morphology	OpenCV (cv2)	HSV histogram, contours, moments
Statistical features	scipy.stats.skew	Channel skewness
Image I/O	Pillow (PIL)	Image loading and resizing
Deployment	Streamlit	Web interface at bacsense.streamlit.app

8. Saved Artifacts

All model artifacts are saved to Google Drive and packaged into bacsense_v2_package.zip for local deployment.

File	Size	Description
vgg16_feature_extractor.keras	~55 MB	Frozen VGG16 feature extractor (no top layer)
pca_model.pkl	~2 MB	PCA model for main pipeline (512 → 94 dims)
standard_scaler.pkl	~0.1 MB	Scaler for main SVM input
svm_classifier.pkl	Variable	Trained 5-class RBF-SVM
class_names.pkl	<1 KB	Ordered list of 5 species names
specialist_svm.pkl	~712 KB	Binary specialist RBF-SVM (prob=True)
specialist_scaler.pkl	~13 KB	Scaler for 683-dim specialist input
specialist_pca.pkl	~1 MB	PCA for specialist (683 → 243 dims)
specialist_metadata.json	<1 KB	Training config, metrics, timestamps
inference.py	~15 KB	Standalone BacSense class for deployment
requirements.txt	<1 KB	Python dependency list

example_usage.py	~3 KB	Quickstart and Streamlit integration code

9. Inference API

9.1 Installation

Install dependencies and point the BacSense class at the unpacked package directory:

Install: pip install tensorflow scikit-learn scikit-image opencv-python scipy Pillow

9.2 Usage

Method	Signature	Returns
Constructor	BacSense(model_dir, specialist_threshold=0.90)	BacSense instance
warmup()	Pre-loads all models into memory	None
predict()	predict(img_path, verbose=False)	Result dict (see below)
predict_batch()	predict_batch(img_paths, verbose=False)	List of result dicts

9.3 Result Dictionary

Key	Type	Example Value
prediction	str	Escherichia coli
confidence	float	0.9621
routed_to_specialist	bool	True
specialist_accepted	bool	True
main_prediction	str	Escherichia coli
gram	str	Negative
shape	str	Rod
risk	str	High

10. Limitations & Future Work

10.1 Current Limitations

- P. aeruginosa recall is 94.80% — approximately 5.2% of P. aeruginosa images are still misclassified.
- The confidence threshold (0.90) was chosen empirically, not calibrated on a held-out validation set.
- The dataset is small (307 original images across 5 classes). Models trained on small datasets may not generalize to different microscopes, staining protocols, or magnifications.
- JPEG augmentation introduces domain shift relative to original PNG images; mitigated by mixing formats in training, but not fully eliminated.
- Inference speed is slower than single-stage models due to sequential feature extraction for routed images.

10.2 Future Work

- Collect 20-30 more original P. aeruginosa images to improve specialist recall beyond 0.95.
- Add Gabor filter features (5 scales x 8 orientations = 40 dims) to further capture P. aeruginosa biofilm texture.
- Calibrate confidence threshold using cross-validated F1 optimization on a dedicated validation set.
- Extend specialist to handle all morphologically similar pairs, not just E. coli / P. aeruginosa.
- Evaluate on images from a different microscopy system to assess generalization.
- Replace binary specialist with a soft-label ensemble combining main SVM and specialist votes.

11. References

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